

SMU Beamer Theme

Your Name

School of Computing and Information Systems
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01

Basic Settings

The SMU Beamer Theme provides a LaTeX beamer template based on SMU's official PowerPoint template, with enhanced features for academic presentations:

- Official SMU branding and colors
- Code highlighting with minted
- Custom block components
- Flexible layout options

Design & Layout:

- Professional SMU branding
- Progress bar footer
- Section transition pages
- Multi-column layouts

Text & Colors:

- Custom color commands
- Text highlighting
- Bold color variants

Code Highlighting:

- Minted integration
- 10+ programming languages
- 49 built-in themes
- Inline code support

Custom Blocks:

- Standard blocks
- Alert blocks
- Mathematical boxes (Theorem, etc.)
- Tagged blocks

Font Styles:

- Regular text in Calibri font
- **Bold text appears in SMU blue**
- *Italic text for emphasis*
- **Important warning** using red bold
- **Key concept** using blue bold

Inline Code Style:

- Use `inline` code for technical terms
- Function names like `printf()` or `main()`
- Styles can be customized in `smucore/smucode.def`

SMU Brand Colors:

- SMU Blue - Primary color
- SMU Light Blue - Secondary color
- SMU Gold - Accent color
- SMU Green - Additional accent

Additional Accent Colors:

- SCIS Yellow - School of Computing
- Alert Red - For warnings
- Muted Blue - Alternative blue
- Muted Green - Alternative green

Inline Mathematics:

Einstein's famous equation $E = mc^2$, Euler's number $e \approx 2.71828$, and pi $\pi \approx 3.14159$.

Display Equations:

$$\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Aligned Equations:

$$\begin{aligned} f(x) &= x^2 + 2x + 1 \\ &= (x + 1)^2 \\ &= (x + 1)(x + 1) \end{aligned}$$

Basic Highlighting:

Normal text with default highlighted content using SMU blue.

Custom Colors:

Different colors: red highlight, green highlight, yellow highlight, and blue highlight.

Math Formula Highlighting:

Inline math: $E = mc^2$ and $a^2 + b^2 = c^2$

Display math:
$$\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

02

Components Design

Standard

This is a standard block with SMU styling.

Alert

Warning block for important information.

Examples

Example block for demonstrating concepts.

Light box for highlighting supplementary information.

Tag

Tagged block for categorized content.

Theorem: Pythagorean Theorem

For a right triangle with sides a and b , and hypotenuse c :

$$a^2 + b^2 = c^2$$

The `\begin{twocolumns}` command defaults to 50% width for each column. You can use `\leftcol[0.6]` and `\rightcol[0.35]` to customize the column widths.

Wider Left Column (60%)

This column takes 60% of the available width, leaving 35% for the right column (plus some spacing).

Useful for:

- Main content vs. sidebar
- Text vs. small figure
- Detailed explanation vs. summary

Narrow Right (35%)

Smaller column for supplementary information.

Step 1

Initialize

Input: Data

Step 2

Process

Transform: Apply

Step 3

Output

Result: Done

Three columns are perfect for showing processes or comparisons.

Example: Performance Comparison

Method	Accuracy (%)	Time (ms)	Memory (MB)
Method A	95.3	12.5	128
Method B	97.1	18.2	256
Method C	96.8	15.7	192
Ours	98.5	14.3	160

Table: Performance comparison of different methods

Use booktabs package for professional-looking tables.

Unordered Lists (Itemize):

- First level item
- Another first level item
 - Second level nested item
 - Another second level item
 - Third level deeply nested
 - More third level content
 - Back to second level
- Back to first level
- Final first level item

Ordered Lists (Enumerate):

1. First step in the process
2. Second step with details
 - 2.1 Sub-step 2.1
 - 2.2 Sub-step 2.2
 - 2.2.1 Sub-sub-step 2.2.1
 - 2.2.2 Sub-sub-step 2.2.2
3. Third step returning to main level
4. Fourth and final step

Custom Description Lists:

- API** Application Programming Interface for software communication
- SDK** Software Development Kit containing tools and libraries
- IDE** Integrated Development Environment for coding
- CI/CD** Continuous Integration and Continuous Deployment

Description lists are perfect for glossaries or term definitions.

03

Code Rendering

The theme uses **minted** for professional code highlighting:

Features:

- Line numbers
- Syntax highlighting
- 49 color themes
- 100+ languages
- Inline code support

Supported Languages:

- Python, C, C++, Java
- JavaScript, TypeScript
- Go, Rust, Shell
- SQL, R, MATLAB
- And many more!

Important

Always use [fragile] option when frames contain code blocks!

```
1 def fibonacci(n):
2     """Calculate the nth Fibonacci number using recursion"""
3     if n <= 1:
4         return n
5     return fibonacci(n-1) + fibonacci(n-2)
6
7 def fibonacci_iterative(n):
8     """Calculate Fibonacci iteratively (more efficient)"""
9     a, b = 0, 1
10    for _ in range(n):
11        a, b = b, a + b
12    return a
13
14    # Test the functions
15    for i in range(10):
16        result = fibonacci_iterative(i)
17        print("F(%d) = %d" % (i, result))
```

Inline Python: `lambda x: x**2`

```

1  #include <iostream>
2  #include <vector>
3  #include <algorithm>
4
5  template<typename T>
6  T findMax(const std::vector<T>& arr) {
7      if (arr.empty()) {
8          throw std::runtime_error("Empty vector");
9      }
10     return *std::max_element(arr.begin(), arr.end());
11 }
12
13 int main() {
14     std::vector<int> numbers = {3, 7, 2, 9, 1, 5};
15     std::cout << "Maximum: " << findMax(numbers) << std::endl;
16
17     std::vector<double> values = {3.14, 2.71, 1.41};
18     std::cout << "Maximum: " << findMax(values) << std::endl;
19
20     return 0;
21 }

```

SQL Query

```

1  SELECT
2      s.student_name,
3      AVG(g.score) as avg_score,
4      COUNT(*) as num_courses
5  FROM students s
6  JOIN grades g
7      ON s.id = g.student_id
8  WHERE g.score >= 80
9  GROUP BY s.id, s.student_name
10 HAVING COUNT(*) >= 3
11 ORDER BY avg_score DESC
12 LIMIT 10;

```

Bash Script

```

1  #!/bin/bash
2
3  BACKUP_DIR="/backup"
4  DATE=$(date +%Y%m%d)
5  SOURCE="/data"
6
7  # Create backup
8  echo "Starting backup..."
9  tar -czf \
10     "$BACKUP_DIR/bk_${DATE}.tar.gz" \
11     "$SOURCE"
12
13  if [ $? -eq 0 ]; then
14     echo "Backup completed"
15  else
16     echo "Backup failed!"
17     exit 1
18  fi

```

Inline code for different languages:

- Python: `def hello(): return "world"`
- C: `int *ptr = malloc(sizeof(int));`
- Java: `List<String> names = new ArrayList<>();`
- JavaScript: `const arr = [1, 2, 3].map(x => x * 2);`

Generic Inline Code

You can also use `\code[language]{code}` for any language:

- Generic: Use `\code[C]printf("Hello")` for output
- Commands: Run `git commit -m "message"`
- Paths: See file `/etc/config.conf`

Getting Started

Load the theme: `\usepackage{smu}`

Code Highlighting

For code blocks: `\begin{frame}[fragile]`

Use environments: `pycode`, `cppcode`, `jscode`, etc.

Columns

Two columns: `\begin{twocolumns}...\end{twocolumns}`

Three columns: `\begin{threecolumns}...\end{threecolumns}`

Documentation

See `README.md` for detailed documentation and examples.

The theme is highly customizable:

Colors Edit `smucore/smucolors.def`

Fonts Edit `smucore/smufonts.def`

Code Style Edit `smucore/smucode.def`

Commands Edit `smucore/smucommands.def`

Blocks Edit `smucore/smublocks.def`

Templates Edit `smucore/smutemplates.def`

All components are modular and can be customized independently!

Documentation

- README.md - Main documentation
- main_test.tex - This demo file
- Source files in smucore/ directory

External Resources

- Minted: <https://ctan.org/pkg/minted>
- Beamer: <https://ctan.org/pkg/beamer>
- Pygments Styles: <https://pygments.org/styles/>

MtA-based Approaches (Multiplicative-to-Additive)

- Paillier Encryption: Requires strong-RSA assumption.
[GGN+16; Lin+17; GG+18; CGG+20; XLX+24]
- CL Encryption: Based on Class Groups.
[CCL+19; CCL+20; YCX+21; DMZ+21]
- Oblivious Transfer: No extra assumptions and computationally efficient.
[DKL+18; DKL+19; DKL+24]
- Joye-Libert: Based on revisited Joye-Libert encryption.
[XAL+23]

Threshold CL Encryption

- Decryption key of CL shared among n parties.
[BDO+23; WMY+23; WMC+24; TX+25; JTX+25]

Others

- Pseudorandom Correlation Generator [ANO+22], ...

- [ANO+22] Damiano Abram, Ariel Nof, Claudio Orlandi, Peter Scholl, and Omer Shlomovits, Low-bandwidth threshold ECDSA via pseudorandom correlation generators. In: *2022 IEEE Symposium on Security and Privacy (SP)*, IEEE, 2022, pp. 2554–2572.
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- [GG+18] Rosario Gennaro and Steven Goldfeder, Fast multiparty threshold ECDSA with fast trustless setup. In: *Proceedings of the 2018 ACM SIGSAC conference on computer and communications security*, 2018, pp. 1179–1194.
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- [TX+25] Guofeng Tang and Haiyang Xue, Robust Threshold ECDSA with Online-Friendly Design in Three Rounds. In: *2025 IEEE Symposium on Security and Privacy (SP)*, IEEE, 2025, pp. 203–221.
- [WMC+24] Harry WH Wong, Jack PK Ma, and Sherman SM Chow, Secure multiparty computation of threshold signatures made more efficient. In: *NDSS*, The Internet Society, 2024.
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Thank You

Q & A